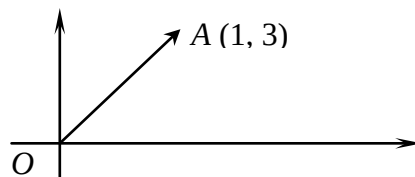


**Question 1.**

**Marks**

- (a) Let  $z_1 = 2 - i$  and  $z_2 = iz_1$ , find  $z_1 - z_2$ . 2
- (b) Let  $z = \frac{-9 + 15i}{4 - i}$ ,
- (i) By simplifying  $\frac{-9 + 15i}{4 - i}$ , express  $z$  in the form  $a + bi$ , 3  
where  $a$  and  $b$  are real numbers.
- (ii) Hence, or otherwise find  $|z|$  and  $\arg z$ . 2
- (c) Solve  $z^2 = 35 - 12i$ , expressing your answer in the form  $a + bi$ . 3
- (d) Sketch the region on an Argand diagram neatly, where the inequalities 4  
 $|z - 2| \leq \operatorname{Re}(z)$  and  $\operatorname{Im}(z) \geq 0$  hold simultaneously.
- (e) Let point  $A$ , below, represent the complex number  $1 + 3i$  in the Argand diagram. 2



*Not to scale*

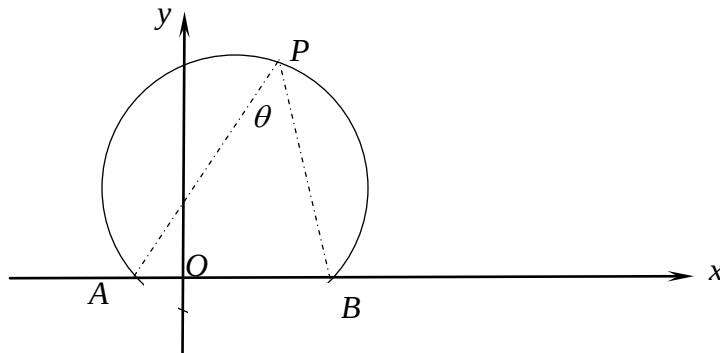
$OA$  is rotated  $45^\circ$  anticlockwise about  $O$  and enlarged by a factor of  $\sqrt{2}$ .  
Find this new complex number in the form  $a + bi$ .

**Question 2. [Start a New Page]**

- (a) Given  $1 - 2i$  is a root of  $x^2 + bx + c = 0$ , where  $b$  and  $c$  are real numbers, 3  
Find the value of  $bc$ .
- (b) (i) Sketch neatly on the same diagram the loci of  $z$  when: 2  
( $\alpha$ )  $|z - (3 + 2i)| = 2$ . ( $\beta$ )  $|z + 3| = |z - 5|$ .  
[clearly label each sketch]
- (ii) Hence write down all the values of  $z$  which satisfy simultaneously: 2  
 $|z - (3 + 2i)| = 2$  and  $|z + 3| = |z - 5|$ .
- (iii) Determine the values of  $k$  for which the simultaneous equations: 2  
 $|z - (3 + 2i)| = 2$  and  $|z - 2i| = k$  have exactly one solution for  $z$ .
- (iv) For the locus of  $z$  when  $|z - (3 + 2i)| = 2$ , 2  
find the maximum value of  $\arg z$ .

- (c) The diagram below shows the locus for all complex numbers  $z$  in the Argand diagram, such that  $\arg\left(\frac{z-3}{z+1}\right) = \frac{\pi}{3}$ .

The angle formed by the intervals  $AP$  and  $BP$  is  $\theta$ , where  $P$  represents the complex number  $z$  and  $A$  is  $(-1, 0)$  and  $B$  is  $(3, 0)$ .



*Not to scale*

- (i) Copy this diagram neatly and find  $\theta$ , giving reasons. 2
- (ii) Find the coordinates for the centre of this circle. 2

### Question 3. [Start a New Page]

- (a) Let  $P$ ,  $Q$  and  $R$  represent the complex numbers  $z_1$ ,  $z_2$  and  $z_3$  respectively. By drawing a diagram or otherwise, determine the geometric properties that characterize  $\triangle PQR$  if  $z_2 - z_1 = i(z_3 - z_1)$ ? Give clear reasons for your answer. 3
- (b) Given  $z = \cos \theta + i \sin \theta$  and that  $m$  and  $n$  are integers.
- (i) Show that  $\frac{z^n + z^{-n}}{2} = \cos n\theta$ . 2
- (ii) Hence, using part (i), show that 2
- $$\cos m\theta \times \cos n\theta = \frac{1}{2} [\cos(m+n)\theta + \cos(m-n)\theta]$$
- (c) Let  $P(z) = z^8 - \frac{5}{2}z^4 + 1$ . The complex number  $\alpha$  is a root of  $P(z) = 0$ .
- (i) Show that  $i\alpha$  and  $\frac{1}{\alpha}$  are also roots of  $P(z) = 0$ . 3
- (ii) Find one of the roots of  $P(z) = 0$  in exact form. 3
- (iii) Hence find all the roots of  $P(z) = 0$ . 2

**Question 4. [Start a New Page]****Marks**

- (a) The point  $P$  on the Argand diagram represents the complex number  $z$ , **3**  
where  $z$  satisfies  $\frac{1}{z} + \frac{1}{\bar{z}} = 1$ .  
Find the locus of  $P$  as  $z$  varies
- (b) (i) State the triangle inequality for the two complex numbers  $z_1$  and  $z_2$ . **1**
- (ii) If  $z_1 = 3 + 4i$  and  $|z_2| = 13$ , find the greatest value for  $|z_1 + z_2|$ . **1**
- (iii) Further, if  $0 < \text{Arg } z_2 < \frac{\pi}{2}$  and  $|z_1 + z_2|$  has this greatest value, **2**  
find  $z_2$ .
- (c) Given  $w = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$ ,
- (i) Show that  $w^k$  is a solution of  $z^5 - 1 = 0$ , where  $k$  is an integer. **2**
- (ii) Explain why  $1 + w + w^2 + w^3 + w^4 = 0$ . **2**
- (iii) Hence, or otherwise find the value of: **2**  
 $(w-1)(1 + 2w + 3w^2 + 4w^3 + 5w^4)$
- (iv) Given that  $S = \frac{w}{1-w^2} + \frac{w^2}{1-w^4} + \frac{w^3}{1-w} + \frac{w^4}{1-w^3}$ , **2**  
Show that  $S = 0$ .

**THE END**

# MATH. EXT 2 SOLUTIONS, TERM 4 2007

## MATHEMATICS Extension 2: Question 1

### Suggested Solutions

### Marks

### Marker's Comments

(a)  $z_1 - z_2 = 2 - i - i(2 - i)$   
 $= 2 - i - 2i - 1$   
 $= 1 - 3i$

[2]

(b) (i)  $z = \frac{(-9 + 15i) \times (4 + i)}{(4 - i)(4 + i)} = \frac{-36 - 9i + 60i - 15}{17}$

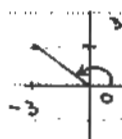
[3]

$z = \frac{-51 + 51i}{17} = -3 + 3i$

(ii)  $|z| = \sqrt{9 + 9} = \sqrt{18} = 3\sqrt{2}$

$\arg z = \frac{3\pi}{4}$

[2]



1 For  $|z| = \sqrt{18} = 3\sqrt{2}$   
 1 For  $\arg z = \frac{3\pi}{4}$

(c) Let  $z = x + iy$ ,  $x, y \in \mathbb{R}$   
 $z^2 = x^2 - y^2 + 2ixy = 35 - 12i$

METHOD 1

$x^2 - y^2 = 35$  — (1)

$xy = -6$  — (2)

so  $x^2y^2 = 36$

and  $x^4 - x^2y^2 = 35x^2$

i.e.  $x^4 - 36 = 35x^2$

$x^4 - 35x^2 - 36 = 0$

$(x^2 - 36)(x^2 + 1) = 0$

$x = \pm 6$  or no real solutions

$\therefore y = \mp 1$

$\therefore z = 6 - i$  or  $-6 + i$

[3]

1 For each solution  
 1 For  $6 - i$   
 1 For  $-6 + i$

(d) Let  $z = x + iy$

$\operatorname{Re}(z) = x$

$|z - z| = \sqrt{(x - x)^2 + y^2}$

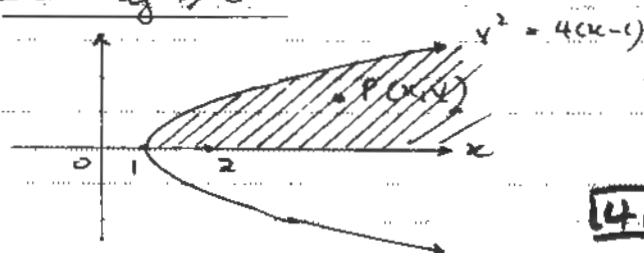
$\operatorname{Im}(z) = y$

$\therefore \sqrt{(x - x)^2 + y^2} \leq x$

$x^2 - 4x + 4 + y^2 \leq x^2$

$y^2 \leq 4(x - 1)$

and  $xy \geq 0$



[4]

$\frac{1}{2}$  each  $\operatorname{Re}(z) = x$   
 $\operatorname{Im}(z) = y$   
 1 For  $\sqrt{(x-x)^2 + y^2} \leq x$   
 1 For  $y^2 \leq 4(x-1)$   
 $\frac{1}{2}$  For sketch of  
 1 For shaded region

(e)

New number  $= (1 + 3i) \times \sqrt{2} \times \operatorname{cis} \frac{\pi}{4}$   
 $= (1 + 3i) \times \sqrt{2} \times \frac{1}{\sqrt{2}} (1 + i)$

[2]

$= -2 + 4i$

# MATHEMATICS Extension 2: Question 2

## Suggested Solutions

Marks

Marker's Comments

(a)  $(1-z)$  is given.  
so  $(1+z)$  is also a root as all the coefficients are real numbers.

1 mk

$$\alpha\beta = \frac{c}{a} = c = (1+z)(1-z) = 5$$

1/2

$$\alpha + \beta = -\frac{b}{a} = -b = 1+2i + 1-2i$$

$$-b = 2$$

$$b = -2$$

1/2

$$\text{so } bc = -2 \times 5 = -10$$

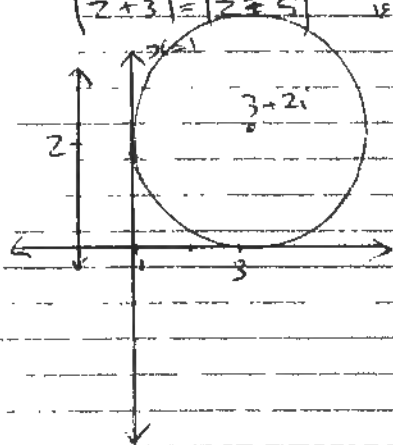
1 mk

(b)(i)  $|z - (3+2i)| = 2$  is a circle centre  $(3+2i)$  radius 2

1/2

$|z+3| = |z-5|$  is the perp. bisector of interval joining  $-3$  and  $5$ , i.e.  $x=1$

1/2 for  $x=1$



1 for circle  
ind. centre

(ii) since  $x=1$  is a tangent to the circle there's only one point of intersection  
 $z = 1+2i$

1 for answer  
1 for explanation

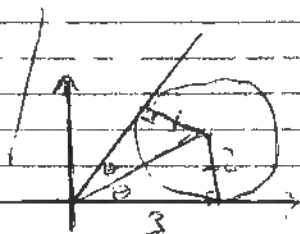
(iii)  $|z-2i|=K$  represents a circle, centre  $2i$ , radius  $K$ .

1 mk for each  
answer



is only one solution for  $z$  when  $K=1$  or  $K=5$

(iv) max arg  $z = 2\theta$   
 $= 2 \tan^{-1}(2/3)$   
 $= 1.1071487$   
 $(63.43^\circ)$



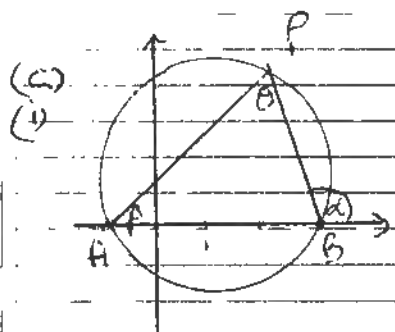
1/2 for diagram/explanation  
1 for  $\theta = \tan^{-1}(2/3)$   
1/2 for answer

# MATHEMATICS Extension 2: Question 2

## Suggested Solutions

## Marks

## Marker's Comments



$$\arg(z-3) = \alpha$$

$$\arg(z+1) = \beta$$

$$\alpha - \beta = \theta$$

(exterior angle of a triangle equals the sum of 2 interior opposite angles)

$$\text{but } \alpha - \beta = \frac{\pi}{3}$$

$$\therefore \theta = \frac{\pi}{3}$$

(ii) centre will lie on  $x=1$ , since the perpendicular bisector of the chord AB passes through the centre



$$\angle AOB = 2\theta$$

$$\angle AOB = \frac{2\pi}{3}$$

$$\frac{y}{2} = \tan \frac{\pi}{6}$$

(angle at centre is double the angle at the circumference)

$$y = \frac{2}{3}\sqrt{3}$$

$\therefore$  centre is  $(1, \frac{2}{3}\sqrt{3})$

1<sup>st</sup> mark

2<sup>nd</sup> mark

1<sup>st</sup> mark

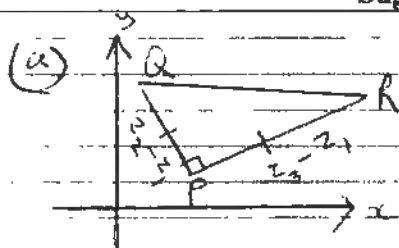
2<sup>nd</sup> mark

# MATHEMATICS Extension 2: Question...3

## Suggested Solutions

## Marks

## Marker's Comments



$\triangle PQR$  is right angled at  $P(z_1)$  and is isosceles  
 $PQ = PR$ .

This is because the multiplication of  $(z_3 - z_1)$  by  $i$  rotates this length  $90^\circ$  anticlockwise, it doesn't change the length as  $|z_2 - z_1| = |i(z_3 - z_1)|$   
 $= |i| \times |z_3 - z_1|$   
 $= |z_3 - z_1|$

1 for answer/diagram

1

1

$$(b) \frac{z^n + z^{-n}}{2} = \frac{(e^{i\theta})^n + (e^{-i\theta})^n}{2}$$

$$= \frac{\cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta)}{2}$$

$$= \frac{\cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta}{2}$$

$$= \frac{2 \cos n\theta}{2}$$

$$= \cos n\theta$$

QED

1 mk

1 mk

$$(ii) \cos(m\theta) \times \cos(n\theta) = \frac{(z^m + z^{-m})}{2} \times \frac{(z^n + z^{-n})}{2}$$

$$= \frac{z^{m+n} + z^{-m-n} + z^{m-n} + z^{-m+n}}{4}$$

$$= \frac{z^{m+n} + z^{-(m+n)} + z^{m-n} + z^{-(m-n)}}{4}$$

$$= \frac{2 \cos(m+n)\theta + 2 \cos(m-n)\theta}{4}$$

$$= \frac{1}{2} [\cos(m+n)\theta + \cos(m-n)\theta]$$

QED

1/2

1/2

1/2

1/2

# MATHEMATICS Extension 2: Question 3

## Suggested Solutions

Marks

Marker's Comments

$$(c)(i) \quad P(z) = z^8 - \frac{5}{2}z^4 + 1$$

$\alpha$  is a root of  $P(z) = 0$

$$\therefore \alpha^8 - \frac{5}{2}\alpha^4 + 1 = 0$$

$$P(i\alpha) = (i\alpha)^8 - \frac{5}{2}(i\alpha)^4 + 1$$

$$= i^8 \alpha^8 - \frac{5}{2}(i^4 \alpha^4) + 1$$

$$= \alpha^8 - \frac{5}{2}\alpha^4 + 1$$

$$= 0$$

$$P\left(\frac{1}{\alpha}\right) = \left(\frac{1}{\alpha}\right)^8 - \frac{5}{2}\left(\frac{1}{\alpha}\right)^4 + 1$$

$$= \frac{1}{\alpha^8} \left(1 - \frac{5}{2}\alpha^4 + \alpha^8\right)$$

$$= \frac{1}{\alpha^8} \times 0$$

$$= 0$$

$\therefore i\alpha$  and  $\frac{1}{\alpha}$  are also roots of  $P(z)$

$$(ii) \quad z^8 - \frac{5}{2}z^4 + 1 = 0$$

$$2z^8 - 5z^4 + 2 = 0$$

$$(2z^4 - 1)(z^4 - 2) = 0$$

$$\therefore z^4 = \frac{1}{2} \text{ or } z^4 = 2$$

one root is  $z = \sqrt[4]{2}$

(iii) the roots are  $\pm\sqrt[4]{2}, \pm\sqrt[4]{2}i, \pm\frac{1}{\sqrt[4]{2}}, \pm\frac{i}{\sqrt[4]{2}}$

2 marks

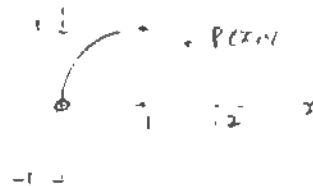
# MATHEMATICS Extension 2: Question 4

## Suggested Solutions

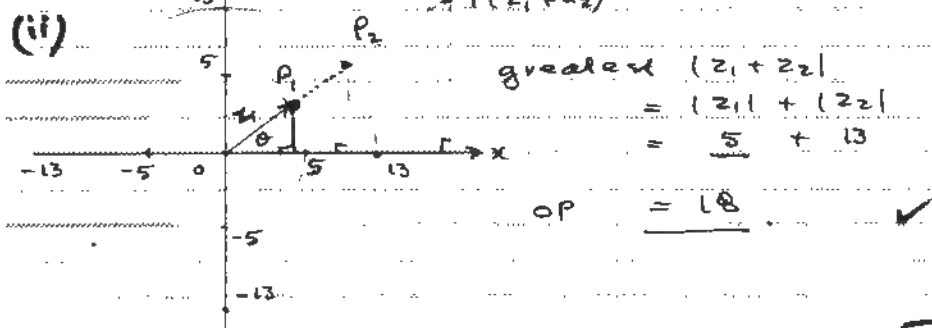
Marks

Marker's Comments

(a) Let  $z = x + iy$  so  $\frac{1}{z} + \frac{1}{\bar{z}} = 1$  [3]  
 $\bar{z} = x - iy$   
 $\frac{z + \bar{z}}{z\bar{z}} = 1$   
 $\frac{2x}{x^2 + y^2} = 1$   
 $\therefore x^2 + y^2 = 2x$   
 $\therefore x^2 - 2x + y^2 = 0$   
 $\therefore (x-1)^2 + y^2 = 1$   
 $\therefore$  locus of  $z$  is  $(x-1)^2 + y^2 = 1$  except  $(0,0)$   
a circle  $c(1,0)$   
 $r=1$



(b) (i)  $|z_1 + z_2| \leq |z_1| + |z_2|$  ✓ equality iff collinear ( $z_2 = kz_1$ )



(iii) As  $z_1$  and  $z_2$  are collinear [2]  
 $OP = 18$   
 $\therefore \vec{OP}_2 = z_2 = \frac{13}{5} \vec{OP}_1 = \frac{13}{5} z_1 = \frac{13}{5} (3+4i)$

$z_2 = 13 \operatorname{cis} \theta$   
where  $\tan \theta = \frac{4}{3}$

(c) (i)  $w = \operatorname{cis} \frac{2\pi}{5}$   
 $[w^k = \operatorname{cis} \frac{2\pi k}{5} \text{ de Moivre's Thm}]$   
 $\therefore z^5 - 1 = (w^k)^5 - 1 = (w^5)^k - 1$   
 $= (\operatorname{cis} \frac{2\pi \times 5}{5})^k - 1$  [2]  
 $= (\operatorname{cis} 2\pi)^k - 1$   
 $= 1 - 1$   
 $\therefore z^5 - 1 = 0$   
 $\therefore w^k$  is a solution of  $z^5 - 1 = 0$   $k \in \mathbb{Z}$

(ii) As  $z^5 - 1 = 0$   $\therefore$  5 roots  $= w^0, w^1, w^2, w^3, w^4$   
 $= 1, w, w^2, w^3, w^4$   
and  $\sum \alpha = 1 + w + w^2 + w^3 + w^4 = -\frac{1}{w} = 0$

[2]

MATHEMATICS Extension 2: Question 4.

Suggested Solutions

Marks

Marker's Comments

(c) (iii)

[2]

$$(w-1)(1+2w+3w^2+4w^3+5w^4)$$

$$= w + 2w^2 + 3w^3 + 4w^4 + 5 \times 1 - 1 - 2w - 3w^2 - 4w^3 - 5w^4$$

$$= -w - w^2 - w^3 - w^4 + 5 - 1$$

$$= -(w + w^2 + w^3 + w^4) + 4$$

$$= -(-1) + 4$$

$$= 5$$

(iv) METHOD 1:  $S = \frac{w}{1-w^2} + \frac{w^2}{1-w^4} + \frac{w^3}{1-w} + \frac{w^4}{1-w^3}$

$$\therefore wS = \frac{w^2}{1-w^2} + \frac{w^3}{1-w^4} + \frac{w^4}{1-w} + \frac{1}{1-w^3}$$

$$= \frac{w^2}{1-w^2} + \frac{w^4}{w-1} + \frac{w^4}{1-w} + \frac{w^2}{w^2-1}$$

[2]  $= \frac{w^2}{1-w^2} + \frac{w^2}{w^2-1} + \frac{w^4}{w-1} + \frac{w^4}{1-w}$

$$= \frac{w^2}{1-w^2} - \frac{w^2}{1-w^2} + \frac{w^4}{w-1} - \frac{w^4}{w-1}$$

$$= 0$$

$$\therefore S = 0 \text{ as } w \neq 0.$$

METHOD 2

$$S = \frac{w}{1-w^2} + \frac{w^2}{1-w^4} + \frac{w^3}{1-w} + \frac{w^4}{1-w^3}$$

$$= \frac{w}{1-w^2} + \frac{w^4}{w^2-w} + \frac{w^4}{w-w^2} + \frac{w}{w^2-1}$$

$$= \frac{w}{1-w^2} - \frac{w}{1-w^2} + \frac{w^4}{w^2-w} - \frac{w^4}{w^2-w}$$

$$= 0$$